

I LLM AND VLM PROMPTING

I.1 LLM PROMPTING

To make the fairest comparison, we use the same prompt style and in-context examples as text2motion[36], which can be found in Appendix B.2. of their paper. We find that having the model generate multiple candidate task-plans during the shooting phase is critical to task performance, so we add a minimal system prompt to make the LLM instruction-following explicit. An example of a full prompt for **LLM-T2M**, $n = 1$ for `hook` `Reach` Task 1 is shown below.

User Prompt
Respond directly in the format specified in the output format section, following the instructions exactly for how many sequences to generate i.e. generate 5 sequences if asked for the "Top 5 robot action sequences". Available primitives: ['pick(a)', 'place(a, b)', 'pull(a, hook)', 'push(a, hook, rack)'] Available predicates: ['on(a, b)', 'inhand(a)', 'under(a, b)'] Available scene objects: ['table', 'blue_box', 'cyan_box', 'hook', 'rack', 'red_box', 'yellow_box'] Object relationships: ['inhand(hook)', 'on(red_box, rack)', 'on(yellow_box, table)', 'on(blue_box, table)', 'on(cyan_box, rack)', 'on(rack, table)'] Human instruction: could you move all the boxes onto the rack? Goal predicate set: [['on(yellow_box, rack)', 'on(blue_box, rack)']] Top 1 robot action sequences: ['pull(yellow_box, hook)', 'place(hook, table)', 'pick(yellow_box)', 'place(yellow_box, rack)', 'pick(blue_box)', 'place(blue_box, rack)'] Available scene objects: ['table', 'rack', 'hook', 'red_box'] Object relationships: ['on(rack, table)', 'on(hook, table)', 'on(red_box, table)'] Human instruction: How would you get the red box in the robot's hand? Goal predicate set: [['inhand(red_box)']] Top 5 robot action sequences (python list of lists):

Model Completion
<pre>[['pick(red_box)'], ['push(red_box, hook, rack)', 'pick(red_box)'], ['pick(hook)', 'place(hook, table)', 'pick(red_box)'], ['pick(hook)', 'place(hook, rack)', 'pick(red_box)'], ['pick(hook)', 'place(hook, table)', 'push(red_box, hook, rack)', 'pick(red_box)']]</pre>

Interestingly, even though many of the prompts contain partial-to-complete solutions in the in-context examples like the one above, we find that this does not guarantee the LLM will generate the correct plan.

A key piece to text2motion's success for planning despite LLM's lack of geometric awareness is the alternation between shooting and greedy search. When the shooting motion planner fails to find a geometrically feasible motion plan from the 5 task plan candidates, the algorithm falls back to a say-can style greedy search that executes the next action deemed most feasible by a combination of the LLM task planner and the motion planner. This allows the LLM task planner to explore with geometric feedback, and is the reason why text2motion is able to solve `Rearrangement Push` Task 1. For our evaluations, we allow 10 tries, which is much more than the steps required to solve any of the tasks in our evaluation suites.

I.2 VLM PROMPTING

For VLM experiments, we modify the system prompt and insert scene images before the scene description. Below is an example of Hook Reach Task 1 with $n = 11$ in-context examples.

User Prompt
Respond directly in the format specified in the output format section, following the instructions exactly for how many sequences to generate i.e. generate 5 sequences if asked for the "Top 5 robot action sequences". Review the provided images carefully when constructing your plan. Available primitives: ['pick(a)', 'place(a, b)', 'pull(a, hook)', 'push(a, hook, rack)'] Available predicates: ['on(a, b)', 'inhand(a)', 'under(a, b)'] Available scene objects: ['table', 'hook', 'rack', 'yellow_box', 'blue_box', 'red_box'] Object relationships: ['inhand(hook)', 'on(yellow_box, table)', 'on(rack, table)', 'on(blue_box, table)'] Human instruction: How would you push two of the boxes to be under the rack? Goal predicate set: [['under(yellow_box, rack)', 'under(blue_box, rack)'], ['under(blue_box, rack)', 'under(red_box, rack)'], ['under(yellow_box, rack)', 'under(red_box, rack)']] Top 1 robot action sequences: ['push(yellow_box, hook, rack)', 'push(red_box, hook, rack)'] Available scene objects: ['table', 'blue_box', 'cyan_box', 'hook', 'rack', 'red_box', 'yellow_box'] Object relationships: ['inhand(hook)', 'on(red_box, rack)', 'on(yellow_box, table)', 'on(blue_box, table)', 'on(cyan_box, rack)', 'on(rack, table)'] Human instruction: could you move all the boxes onto the rack? Goal predicate set: [['on(yellow_box, rack)', 'on(blue_box, rack)']] Top 1 robot action sequences: ['pull(yellow_box, hook)', 'place(hook, table)', 'pick(yellow_box)', 'place(yellow_box, rack)', 'pick(blue_box)', 'place(blue_box, rack)'] Available scene objects: ['table', 'blue_box', 'hook', 'rack', 'red_box', 'yellow_box'] Object relationships: ['on(hook, table)', 'on(red_box, table)', 'on(blue_box, table)', 'on(yellow_box, rack)', 'on(rack, table)'] Human instruction: Move the ocean colored box to be under the rack and ensure the hook ends up on the table. Goal predicate set: [['under(blue_box, rack)']] Top 1 robot action sequences: ['pick(red_box)', 'place(red_box, table)', 'pick(yellow_box)', 'place(yellow_box, rack)', 'pick(hook)', 'push(blue_box, hook, rack)', 'place(hook, table)'] Available scene objects: ['table', 'cyan_box', 'hook', 'red_box', 'yellow_box', 'rack', 'blue_box'] Object relationships: ['on(hook, table)', 'on(red_box, table)', 'on(blue_box, table)', 'on(cyan_box, table)', 'on(rack, table)', 'under(yellow_box, rack)'] Human instruction: How would you get the cyan box under the rack and then ensure the hook is on the table? Goal predicate set: [['under(cyan_box, rack)', 'on(hook, table)']] Top 1 robot action sequences: ['pick(blue_box)', 'place(blue_box, table)', 'pick(red_box)', 'place(red_box, table)', 'pick(hook)', 'push(cyan_box, hook, rack)', 'place(hook, table)']

Interestingly, we find that including images in the prompt degrades the performance.

User Prompt

Available scene objects: ['table', 'cyan_box', 'hook', 'blue_box', 'rack', 'red_box']
 Object relationships: ['on(hook, table)', 'on(rack, table)', 'on(blue_box, table)', 'on(cyan_box, table)', 'on(red_box, table)']
 Human instruction: How would you push all the boxes under the rack? Goal predicate set: [['under(blue_box, rack)', 'under(cyan_box, rack)', 'under(red_box, rack)']]
 Top 1 robot action sequences: ['pick(blue_box)', 'place(blue_box, table)', 'pick(hook)', 'push(cyan_box, hook, rack)', 'place(hook, table)', 'pick(blue_box)', 'place(blue_box, table)', 'pick(hook)', 'push(blue_box, hook, rack)', 'push(red_box, hook, rack)']

Available scene objects: ['table', 'cyan_box', 'hook', 'rack', 'red_box', 'blue_box']
 Object relationships: ['on(hook, table)', 'on(cyan_box, rack)', 'on(rack, table)', 'on(red_box, table)', 'inhand(blue_box)']
 Human instruction: How would you set the red box to be the only box on the rack?
 Goal predicate set: [['on(red_box, rack)', 'on(blue_box, table)', 'on(cyan_box, table)']]
 Top 1 robot action sequences: ['place(blue_box, table)', 'pick(hook)', 'pull(red_box, hook)', 'place(hook, table)', 'pick(red_box)', 'place(red_box, rack)', 'pick(cyan_box)', 'place(cyan_box, table)']

Available scene objects: ['table', 'cyan_box', 'red_box', 'hook', 'rack']
 Object relationships: ['on(hook, table)', 'on(rack, table)', 'on(cyan_box, rack)', 'on(red_box, rack)']
 Human instruction: put the hook on the rack and stack the cyan box above the rack - thanks
 Goal predicate set: [['on(hook, rack)', 'on(cyan_box, rack)']]
 Top 1 robot action sequences: ['pick(hook)', 'pull(cyan_box, hook)', 'place(hook, rack)', 'pick(cyan_box)', 'place(cyan_box, rack)']

Available scene objects: ['table', 'cyan_box', 'hook', 'rack', 'red_box', 'blue_box']
 Object relationships: ['on(hook, table)', 'on(blue_box, rack)', 'on(cyan_box, table)', 'on(red_box, table)', 'on(rack, table)']
 Human instruction: Move the warm colored box to be underneath the rack.
 Goal predicate set: [['under(red_box, rack)']]
 Top 1 robot action sequences: ['pick(blue_box)', 'place(blue_box, table)', 'pick(red_box)', 'place(red_box, table)', 'pick(hook)', 'push(red_box, hook, rack)']

Available scene objects: ['table', 'blue_box', 'red_box', 'hook', 'rack', 'yellow_box']
 Object relationships: ['on(hook, table)', 'on(blue_box, table)', 'on(rack, table)', 'on(red_box, table)', 'on(yellow_box, table)']
 Human instruction: situate an odd number greater than 1 of the boxes above the rack
 Goal predicate set: [['on(blue_box, rack)', 'on(red_box, rack)', 'on(yellow_box, rack)']]
 Top 1 robot action sequences: ['pick(hook)', 'pull(blue_box, hook)', 'place(hook, table)', 'pick(blue_box)', 'place(blue_box, rack)', 'pick(red_box)', 'place(red_box, rack)', 'pick(yellow_box)', 'place(yellow_box, rack)']

User Prompt

continued...

Available scene objects: ['table', 'cyan_box', 'hook', 'yellow_box', 'blue_box', 'rack']

Object relationships: ['on(hook, table)', 'on(yellow_box, rack)', 'on(rack, table)', 'on(cyan_box, rack)']

Human instruction: set the hook on the rack and stack the yellow box onto the table and set the cyan box on the rack

Goal predicate set: [['on(hook, rack)', 'on(yellow_box, table)', 'on(cyan_box, rack)']]

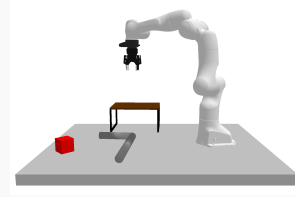
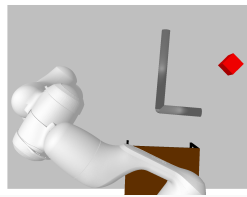
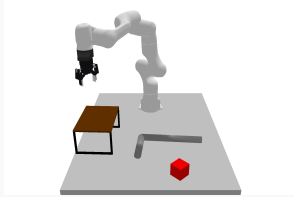
Top 1 robot action sequences: ['pick(yellow_box)', 'place(yellow_box, table)', 'pick(hook)', 'pull(yellow_box, hook)', 'place(hook, table)']

Available scene objects: ['table', 'rack', 'hook', 'cyan_box', 'yellow_box', 'red_box']

Object relationships: ['on(yellow_box, table)', 'on(rack, table)', 'on(cyan_box, table)', 'on(hook, table)', 'on(red_box, rack)']

Human instruction: Pick up any box.

Goal predicate set: [['inhand(yellow_box)'], ['inhand(cyan_box)']] Top 1 robot action sequences: ['pick(yellow_box)']



Available scene objects: ['table', 'rack', 'hook', 'red_box']

Object relationships: ['on(rack, table)', 'on(hook, table)', 'on(red_box, table)']

Human instruction: How would you get the red box in the robot's hand? Goal predicate set: [['inhand(red_box)']]

Top 5 robot action sequences (python list of lists):

Model Completion

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[ ['pick(red_box)'], ['pick(hook)', 'place(hook, table)', 'pick(red_box)'], ['pick(rack)', 'place(rack, table)', 'pick(red_box)'], ['pick(hook)', 'place(hook, table)', 'pick(rack)', 'place(rack, table)', 'pick(red_box)'], ['pick(red_box)', 'place(red_box, table)', 'pick(red_box)'] ]
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